

Mikel Yu

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EDUCATION

- **Technical University of Denmark (DTU)** *Expected: Feb 2027 – Dec 2028*
Master of Science in Human-Centered Artificial Intelligence GPA: - / 12.0
- **Technical University of Denmark (DTU)** *Sep 2023 – Jun 2026*
Bachelor of Science in Artificial Intelligence and Data GPA: 9.0 / 12.0
- **Johns Hopkins University (JHU)** *Sep 2025 – Dec 2025*
Exchange Semester GPA: 3.70 / 4.00

PUBLICATIONS

- **Yu, Mikel, and Fernando, Benjamin - In review at Acta Astronautica | Preprint**  *2026*
Seismoacoustic recovery of the SpaceX CRS-32 Dragon re-entry with a proof-of-concept for automated detection
 - Built directly on Dr. Fernando's Science-published work; implemented end-to-end pipeline to detect and localize atmospheric re-entries using open-source seismic network data. Constructed a gradient-based contour mapping and predictive line fitting for trajectory triangulation.
 - Fine-tuned a classification model on limited labelled data, achieving high accuracy on held-out examples, showing promise for ML to automate seismic phase picking
- **Yu, Mikel, et al. - Manuscript in preparation**  *Feb 2026 – Jun 2026*
B.Sc. Thesis: Counterfactual Explanations for ML Triage Lyngby, Denmark
 - Developed a counterfactual explanation framework for a deployed ML-based patient acuity classification model, enabling causal what-if queries over the feature space of a live triage system.
 - Graded 12/12; both supervisor and external examiner assessed work as publication-worthy

EXPERIENCE

- **Johns Hopkins University** *Sep 2025 – Dec 2025*
Independent Undergraduate Researcher, Morton K. Blaustein Department of Earth and Planetary Sciences Maryland, USA
 - Conducted independent research under the supervision of Reines Distinguished Fellow Dr. Fernando, building on his Science-published work on atmospheric re-entry events. Validated atmospheric re-entry detection methodologies, extending prior Science-published research; developed automated ML pipelines for seismic signal phase picking.
- **Technical University of Denmark** *Jan 2025 – Jan 2026*
Teaching Assistant, Computer Science and Artificial Intelligence Lyngby, Denmark
 - Taught Foundational Mathematics and facilitated daily problem-solving sessions and proctored digital assessments across two course iterations. Curriculum: calculus, complex numbers, linear algebra, differential equations
- **TopTutors** *Aug 2023 – Present*
Private Tutor, Mathematics and other high school subjects Remote
 - Ongoing tutoring of high school and middle school students in various subjects; primary focus on mathematics at calculus level

PROJECTS

- **Adversarialness of Adversarial attacks** 
Quantification and benchmarking of adversarial attacks
 - Benchmarked and quantified several performance metrics of adversarial attacks. Additionally, proposed a novel true class robustness ratio (TCRR) metric to quantify model confidence degradation in the target class

AWARDS AND ACHIEVEMENTS

- **Anthropic Fellows Program — AI Safety, Anthropic, 2026:** Selected as a technical qualifier through multiple rigorous technical rounds in a globally competitive fellowship.
- **Merit Award - High School Valedictorian, Hasseris Gymnasium, 2022:** Graduated from with the highest GPA in the entire year, achieving a grade of 12.4/12.0 by undertaking two additional A-level subjects.

TECHNICAL SKILLS

Programming: Python, Java, R, C

Machine learning: PyTorch, TensorFlow, scikit-learn, NumPy, pandas

HPC: IBM Spectrum LSF, Linux, Git